

Homework 2: Probability

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Problem 1

- (a) Prove from the axioms that if $Y \subseteq Z$, then $p(Y) \leq p(Z)$.

Proof. Since $Y \subseteq Z$, we can write $Z = Y \cup (Z - Y)$ with a disjoint union. Then by additivity property given,

$$p(Z) = p(Y) + p(Z - Y) \geq p(Y).$$

□

- (b) Use the above fact to prove that conditional probabilities $p(X | Z)$, like ordinary probabilities, always fall in $[0, 1]$.

Proof. If $p(Z) = 0$, then $p(X | Z)$ is undefined. If $p(Z) > 0$, then $0 \leq p(X \cap Z) \leq p(Z)$, hence

$$0 \leq \frac{p(X \cap Z)}{p(Z)} \leq 1.$$

□

- (c) Prove from the axioms that $p(\emptyset) = 0$.

Proof. Using disjoint additivity with $E = E \cup \emptyset$,

$$p(E) = p(E) + p(\emptyset) \Rightarrow p(\emptyset) = 0.$$

□

- (d) Let $\bar{X} := E - X$. Prove that $p(X) = 1 - p(\bar{X})$.

Proof. Since $E = X \cup \bar{X}$ and the union is disjoint,

$$1 = p(E) = p(X) + p(\bar{X}) \Rightarrow p(X) = 1 - p(\bar{X}).$$

□

- (e) Prove that $p(\text{singing AND rainy} | \text{rainy}) = p(\text{singing} | \text{rainy})$.

Proof. Let s be the event “singing” and r be “rainy”. Then

$$p(s \cap r | r) = \frac{p(s \cap r \cap r)}{p(r)} = \frac{p(s \cap r)}{p(r)} = p(s | r).$$

□

- (f) Prove that $p(X | Y) = 1 - p(\bar{X} | Y)$.

Proof. Note $Y = (X \cap Y) \cup (\bar{X} \cap Y)$ with disjoint union. Hence

$$p(Y) = p(X \cap Y) + p(\bar{X} \cap Y).$$

Dividing by $p(Y)$ (assuming $p(Y) > 0$) gives

$$1 = p(X | Y) + p(\bar{X} | Y) \Rightarrow p(X | Y) = 1 - p(\bar{X} | Y).$$

□

(g) Simplify:

$$(p(X | Y)p(Y) + p(X | \bar{Y})p(\bar{Y})) \cdot \frac{p(\bar{Z} | X)}{p(\bar{Z})}.$$

Answer We start from the given expression

$$(p(X | Y)p(Y) + p(X | \bar{Y})p(\bar{Y})) \cdot \frac{p(\bar{Z} | X)}{p(\bar{Z})}.$$

Expand the conditional probabilities: $p(X | Y) = \frac{p(X \cap Y)}{p(Y)}$, $p(X | \bar{Y}) = \frac{p(X \cap \bar{Y})}{p(\bar{Y})}$

Hence,

$$p(X | Y)p(Y) + p(X | \bar{Y})p(\bar{Y}) = p(X \cap Y) + p(X \cap \bar{Y}).$$

Since Y and $\bar{Y}$ form a partition, we have

$$(X \cap Y) \cap (X \cap \bar{Y}) = \emptyset, \quad (X \cap Y) \cup (X \cap \bar{Y}) = X,$$

so

$$p(X \cap Y) + p(X \cap \bar{Y}) = p((X \cap Y) \cup (X \cap \bar{Y})) = p(X).$$

Therefore, the whole expression reduces to

$$p(X) \cdot \frac{p(\bar{Z} | X)}{p(\bar{Z})} = p(X) \cdot \frac{\frac{p(X \cap \bar{Z})}{p(X)}}{p(\bar{Z})} = \frac{p(X \cap \bar{Z})}{p(\bar{Z})} = \boxed{p(X | \bar{Z})}.$$

(h) Under what conditions is it true that $p(\text{singing OR rainy}) = p(\text{singing}) + p(\text{rainy})$?

Proof. By the addition rule,

$$p(\text{singing OR rainy}) = p(\text{singing}) + p(\text{rainy}) - p(\text{singing} \cap \text{rainy}).$$

So the equality holds exactly when

$$p(\text{singing} \cap \text{rainy}) = 0.$$

□

(i) Under what conditions is it true that $p(\text{singing AND rainy}) = p(\text{singing}) \cdot p(\text{rainy})$?

Proof. By the definition of conditional probability,

$$p(\text{singing AND rainy}) = p(\text{singing} | \text{rainy}) p(\text{rainy}).$$

So the equality holds exactly when

$$p(\text{singing} | \text{rainy}) = p(\text{singing}),$$

i.e. the probability of singing does not change whether or not it is rainy (say, independent). □

(j) Suppose $p(X | Y) = 0$. Prove that $p(X | Y, Z) = 0$.

Proof. From $p(X | Y) = 0$ (with $p(Y) > 0$) we have $p(X \cap Y) = 0$. Since $X \cap Y \cap Z \subseteq X \cap Y$, we get $p(X \cap Y \cap Z) = 0$. Hence (if $p(Y \cap Z) > 0$),

$$p(X | Y, Z) = \frac{p(X \cap Y \cap Z)}{p(Y \cap Z)} = 0,$$

or else it is undefined in the case of $0/0$. □

(k) Suppose $p(W | Y) = 1$. Prove that $p(W | Y, Z) = 1$.

Proof. $p(W | Y) = 1$ (with $p(Y) > 0$) implies $p(W \cap Y) = p(Y)$, i.e., $Y \subseteq W$. Then $W \cap Y \cap Z = Y \cap Z$, so (if $p(Y \cap Z) > 0$),

$$p(W | Y, Z) = \frac{p(W \cap Y \cap Z)}{p(Y \cap Z)} = 1,$$

or else undefined ($0/0$). □

Problem 2

All cars are either red or blue. The witness claimed the car that hit the pedestrian was blue. Witnesses are believed to be about 80% reliable in reporting car color, regardless of the actual car color. But only 10% of all cars are blue.

- (a) Write an equation relating the following quantities and perhaps other quantities:

$$\begin{aligned} p(\text{Actual} = \text{blue}) \\ p(\text{Actual} = \text{blue} \mid \text{Claimed} = \text{blue}) \\ p(\text{Claimed} = \text{blue} \mid \text{Actual} = \text{blue}). \end{aligned}$$

Answer By Bayes' Theorem,

$$p(\text{Actual} = \text{blue} \mid \text{Claimed} = \text{blue}) = \frac{p(\text{Claimed} = \text{blue} \mid \text{Actual} = \text{blue}) p(\text{Actual} = \text{blue})}{p(\text{Claimed} = \text{blue})}.$$

- (b) Match the events $\text{Actual} = \text{blue}$ and $\text{Claimed} = \text{blue}$ with the following terms:

the hypothesis you are evaluating
the evidence you have observed.

Similarly, match the 3 probabilities in question 2(a) with the following terms:

prior probability of the hypothesis
likelihood of the hypothesis
posterior probability of the hypothesis.

Answer:

Event mapping: $\text{Actual} = \text{blue}$ is *hypothesis*; $\text{Claimed} = \text{blue}$ is *evidence*.

Probability mapping: $p(\text{Actual} = \text{blue}) = \text{prior}$; $p(\text{Claimed} = \text{blue} \mid \text{Actual} = \text{blue}) = \text{likelihood}$; $p(\text{Actual} = \text{blue} \mid \text{Claimed} = \text{blue}) = \text{posterior}$.

- (c) Give the values of all three probabilities in question 2(a). (Hint: Use Bayes' Theorem.) Which probability should the judge care about?

Answer:

From the problem statement: **prior** $p(\text{Actual} = \text{blue}) = 0.1$, $p(\text{Actual} = \text{red}) = 0.9$.

Witness reliability is 80% regardless of color, so

$$p(\text{Claimed} = \text{blue} \mid \text{Actual} = \text{blue}) = 0.8, \quad p(\text{Claimed} = \text{blue} \mid \text{Actual} = \text{red}) = 0.2.$$

Hence

$$\begin{aligned} p(\text{Claimed} = \text{blue}) &= p(\text{Claimed} = \text{blue AND Actual} = \text{blue}) \\ &\quad + p(\text{Claimed} = \text{blue AND Actual} = \text{red}) \\ &= p(\text{Claimed} = \text{blue} \mid \text{Actual} = \text{blue})p(\text{Actual} = \text{blue}) \\ &\quad + p(\text{Claimed} = \text{blue} \mid \text{Actual} = \text{red})p(\text{Actual} = \text{red}) \\ &= 0.8 \cdot 0.1 + 0.2 \cdot 0.9 \\ &= 0.08 + 0.18 \\ &= 0.26. \end{aligned}$$

By Bayes:

$$p(\text{Actual} = \text{blue} \mid \text{Claimed} = \text{blue}) = \frac{0.8 \cdot 0.1}{0.26} \approx 0.3077.$$

So the three requested values are:

$$\begin{aligned} \textbf{prior:} \quad & p(\text{Actual} = \text{blue}) = 0.1 \\ \textbf{likelihood:} \quad & p(\text{Claimed} = \text{blue} \mid \text{Actual} = \text{blue}) = 0.8 \\ \textbf{posterior:} \quad & p(\text{Actual} = \text{blue} \mid \text{Claimed} = \text{blue}) = 4/13 \approx 0.3077 \end{aligned}$$

The judge should care about the *posterior* $p(\text{Actual} = \text{blue} \mid \text{Claimed} = \text{blue})$.

- (d) Let's suppose the numbers 80% and 10% are specific to Baltimore. So in the previous problem, you were implicitly using the following more general version of Bayes' Theorem:

$$p(A | B, Y) = \frac{p(B | A, Y) p(A | Y)}{p(B | Y)},$$

where Y is city = Baltimore. Carefully prove it.

Proof. By the definition of conditional probability, $p(A | B, Y) = \frac{p(A \cap B | Y)}{p(B | Y)}$. Also $p(A \cap B | Y) = p(B | A, Y) p(A | Y)$ by the product rule under the background condition Y . Substitute to obtain

$$p(A | B, Y) = \frac{p(B | A, Y) p(A | Y)}{p(B | Y)}.$$

□

- (e) Now prove the more detailed version

$$p(A | B, Y) = \frac{p(B | A, Y) p(A | Y)}{p(B | A, Y) p(A | Y) + p(B | \bar{A}, Y) p(\bar{A} | Y)}.$$

Proof. By the law of total probability under Y ,

$$p(B | Y) = p(B | A, Y) p(A | Y) + p(B | \bar{A}, Y) p(\bar{A} | Y).$$

Plug this denominator into the result from part (d) to obtain the stated fraction.

□

- (f) Write out the equation given in question 2(e) with A, B and Y replaced by specific propositions from the red-and-blue car problem. For example, Y is "city= Baltimore" (or just "Baltimore" for short). Now replace the probabilities with actual numbers from the problem, such as 0.8. Yeah, it's a mickeymouse problem, but I promise that writing out a real case of this important formula won't kill you, and may even be good for you (like, on an exam).

Answer Let A be the event 'Actual', C be the event 'Claimed', Y be the event 'City'. Then $p(A = \text{blue} | C = \text{blue}, \text{Baltimore}) =$

$$\frac{p(C = \text{blue} | A = \text{blue}, \text{Baltimore}) p(A = \text{blue} | \text{Baltimore})}{p(C = \text{blue} | A = \text{blue}, \text{Baltimore}) p(A = \text{blue} | \text{Baltimore}) + p(C = \text{blue} | A = \text{red}, \text{Baltimore}) p(A = \text{red} | \text{Baltimore})}.$$

which equals

$$\frac{0.8 \cdot 0.1}{0.8 \cdot 0.1 + 0.2 \cdot 0.9} = \frac{4}{13} \approx 0.3077.$$

Problem 3

Leave out the problem descriptions.

$p(\text{cry} \text{situation})$	Predator!	Timber!	I need help!
bwa	0	0.1	0.8
bwee	0	0.6	0.1
kiki	1.0	0.3	0.1

- (a) **Answer.**

$$\sum_{\text{cry} \in \{\text{bwa}, \text{bwee}, \text{kiki}\}} p(\text{cry} | \text{situation} = s) = 1 \quad \text{for each } s \in \{\text{Predator!}, \text{Timber!}, \text{I need help!}\}.$$

- (b) $p(\text{timber}) = 0$ and $p(\text{predator}) = 0.2$. Fill in the rest of the table, using the previous table and the laws of probability.
Answer. Since the three situations form a partition, $p(\text{I need help}) = 1 - p(\text{Predator}) - p(\text{Timber}) = 1 - 0.2 - 0 = 0.8$.
 By $p(\text{cry}, \text{situation}) = p(\text{cry} | \text{situation}) p(\text{situation})$, the joint table is

$p(\text{cry}, \text{situation})$	Predator!	Timber!	I need help!	TOTAL
bwa	$0 \cdot 0.2 = 0$	$0.1 \cdot 0 = 0$	$0.8 \cdot 0.8 = \mathbf{0.64}$	0.64
bwee	$0 \cdot 0.2 = 0$	$0.6 \cdot 0 = 0$	$0.1 \cdot 0.8 = \mathbf{0.08}$	0.08
kiki	$1.0 \cdot 0.2 = \mathbf{0.20}$	$0.3 \cdot 0 = 0$	$0.1 \cdot 0.8 = \mathbf{0.08}$	0.28
TOTAL	0.2	0	0.8	1

- (c) A beaver in this colony cries *kiki*. Given this cry, other beavers try to figure out the probability that there is a predator.

(i) **Answer.** $p(\text{Predator} \mid \text{kiki})$.

(ii) **Answer.** Without the $\mid$ symbol: $p(\text{Predator} \mid \text{kiki}) = \frac{p(\text{Predator} \cap \text{kiki})}{p(\text{kiki})}$.

(iii) **Answer.** Using the joint table, the value is:

$$\frac{p(\text{kiki}, \text{Predator})}{p(\text{kiki})} = \frac{0.20}{0.28} \approx 0.7142857.$$

(iv) **Answer.** Alternatively, Bayes' Theorem allows to express as:

$$p(\text{Predator} \mid \text{kiki}) = \frac{p(\text{kiki} \mid \text{Predator}) p(\text{Predator})}{p(\text{kiki} \mid \text{Predator}) p(\text{Predator}) + p(\text{kiki} \mid \text{Timber}) p(\text{Timber}) + p(\text{kiki} \mid \text{Help}) p(\text{Help})}.$$

(v) **Answer.** The value is:

$$\frac{1.0 \cdot 0.2}{1.0 \cdot 0.2 + 0.3 \cdot 0 + 0.1 \cdot 0.8} = \frac{0.2}{0.2 + 0 + 0.08} = \frac{5}{7}.$$

Problem 4

Leave out the problem descriptions.

(a) **Answer.**

$$\begin{aligned} p(w_1 w_2 w_3 w_4) &= p(w_1 \mid \text{BOS}, \text{BOS}) p(w_2 \mid \text{BOS}, w_1) p(w_3 \mid w_1, w_2) p(w_4 \mid w_2, w_3) p(\text{EOS} \mid w_3, w_4) \\ &= \frac{c(\text{BOS BOS } w_1)}{c(\text{BOS BOS})} \frac{c(\text{BOS } w_1 w_2)}{c(\text{BOS } w_1)} \frac{c(w_1 w_2 w_3)}{c(w_1 w_2)} \frac{c(w_2 w_3 w_4)}{c(w_2 w_3)} \frac{c(w_3 w_4 \text{ EOS})}{c(w_3 w_4)}. \end{aligned}$$

Within which,

- $c(\text{BOS BOS})$ = count of the bigram BOS BOS, which can be estimated by the number of sentences in the whole corpus.
- $c(\text{BOS BOS } i)$ = number of sentences that begin with the word i .
- $c(\text{bagpipe music EOS})$ = number of occurrences where sentences ending with “bagpipe music”.

- (b) **Answer.** Because valid English sentences never end with the determiner “the,” so the factor $p(\text{EOS} \mid \text{think, the})$ is tiny (near 0, because $c(\text{think the EOS})$ is very small).
- (c) Match up expressions (A), (B), (C) with descriptions (1), (2), (3):

- (A) $p(\text{do}) \cdot p(\text{you} \mid \text{do}) \cdot p(\text{think} \mid \text{do, you})$
 (B) $p(\text{do} \mid \text{BOS}) \cdot p(\text{you} \mid \text{BOS, do}) \cdot p(\text{think} \mid \text{do, you}) \cdot p(\text{EOS} \mid \text{you, think})$
 (C) $p(\text{do} \mid \text{BOS}) \cdot p(\text{you} \mid \text{BOS, do}) \cdot p(\text{think} \mid \text{do, you})$

- (1) the first complete sentence you hear is *do you think*
 (2) the first 3 words you hear are *do you think*, all as part of a single sentence
 (3) the first complete sentence you hear starts with *do you think*

Explain your answers briefly.

Answer. (B) $\leftrightarrow$ (1): (1) states that I hear a **complete** sentence. So the representation includes BOS at the start and EOS after “think”.

(C) $\leftrightarrow$ (3): includes BOS but not EOS, so it is the probability that a complete sentence starts with “do you think” (and may continue).

(A) $\leftrightarrow$ (2): omits BOS/EOS, so it is the probability that the first three words you hear are “do you think” within a single sentence (not necessarily at sentence start or end).

(d) *Proof.* We unfold the expression for both models, based on $p(c \mid a, b) = \frac{c(abc)}{c(ab)}$. Thus

$$p_{\text{sequential}}(\vec{w}) = \prod_{i=1}^{n+1} \frac{c(w_{i-2} w_{i-1} w_i)}{c(w_{i-2} w_{i-1})} = \frac{\prod_{i=1}^{n+1} c(w_{i-2} w_{i-1} w_i)}{\prod_{i=1}^{n+1} c(w_{i-2} w_{i-1})}.$$

Similarly, for the reversed model trained on the *same* corpus, $p(w_i \mid w_{i+1}, w_{i+2}) = \frac{c(w_i w_{i+1} w_{i+2})}{c(w_{i+1} w_{i+2})}$, so

$$p_{\text{reversed}}(\vec{w}) = \prod_{i=0}^n \frac{c(w_i w_{i+1} w_{i+2})}{c(w_{i+1} w_{i+2})} = \frac{\prod_{i=0}^n c(w_i w_{i+1} w_{i+2})}{\prod_{i=0}^n c(w_{i+1} w_{i+2})}.$$

Noting the index shifts and conventions $w_{-1} = w_0 = \text{BOS}$ and $w_{n+1} = w_{n+2} = \text{EOS}$, the numerator sets $\{c(w_{i-2}w_{i-1}w_i) : i = 1, \dots, n+1\}$ and $\{c(w_iw_{i+1}w_{i+2}) : i = 0, \dots, n\}$ are the same multiset of trigram counts from the training corpus (each interior trigram appears exactly once in each product, and the boundary trigrams BOS BOS w_1 and $w_{n-1} w_n$ EOS appear as well). In other words, there exists a one-to-one corresponding among the factors within two products. For example, $c_{\text{sequential}}(ABC) = c_{\text{reverse}}(CBA)$ for sure.

So does the denominator products.

Therefore the two fractions are equal, yielding $p(\vec{w}) = p_{\text{reversed}}(\vec{w})$. □

Problem 5

Answer. Introduce a latent variable $a \in \{1, \dots, k\}$ covering topic information. The topic is chosen once per sentence and then persists throughout the sentence. Then, conditioning on a ,

$$p(w_1w_2w_3w_4) = \sum_a p(a) p(w_1 \mid \text{BOS}, a) p(w_2 \mid w_1, a) p(w_3 \mid w_2, a) p(w_4 \mid w_3, a)$$

Problem 6

Answer. Hope I can do well in the quiz

Problem 7

Answer. See Zimo Qi's post on piazza.

Problem 8

(a) **Answer.**

```
python findsim.py words-50.txt <word>
```

- seattle: seahawks dallas atlanta wichita tacoma lauderdale florida spokane chino dulles
- dog: dogs badger cat hound puppy dachshund sighthound poodle rat keeshond
- communist: socialist bolshevik communists comintern trotskyist leftist bolsheviks cominform stalinis
leninist
- jpg: png svg szczepanek buteo pix gif image galleria regnum fiav
- the: in its of which entire within from a part second
- google: webpage yahoo web search com blogging faq editable archived redirection

```
python findsim.py words-20.txt <word>
```

- seattle: atlanta dallas miami canucks indianapolis cincinnati lauderdale florida knicks pensacola
- dog: cat dogs ass badger hound sighthound canine azawakh bikini quadruped
- communist: socialist communists comintern fascist trotskyist cnt agitating bolshevik nationalist
leninist
- jpg: png colspan ffffff bratislava verbena gallery hoek alster hopewell beacon
- the: of in within its between entire over part uninterrupted marked
- google: geocaching portals info web yahoo directory dmoz com wiki search

```
python findsim.py words-100.txt <word>
```

- seattle: seahawks tacoma spokane redskins intelligencer expos miami kahului baltimore atlanta
- dog: dogs hound badger inu komondor mastiff canine borzoi puppy keeshond
- communist: socialist comintern bolshevik communists leftist rightist marxist communism trotskyist
stalin
- jpg: png svg caption gif alpina mjh modis regnum daressalaam szczepanek
- the: of its in which entire a itself this second from
- google: yahoo gmail msn archived search pagerank skype webpage lycos maps

Discuss

For the 50-dimensional embeddings:

- **seattle:**
Good neighbors: seahawks, tacoma, spokane, dallas, atlanta, which are cities or local entities.
Poor neighbors: chino, lauderdale, florida, dulles, which are unrelated or very distant geographically.
- **dog:**
Good neighbors: dogs, puppy, hound, dachshund, poodle, sighthound, keeshond, all directly related to dogs.
Poor neighbors: badger, rat, which are animals but not dogs.
- **communist:**
Good neighbors: socialist, bolshevik, comintern, trotskyist, stalinist, leninist, all strongly related ideologically.
Poor neighbors: duplicates such as communists, bolsheviks, showing weak handling of morphology. Semantically speaking, most of its neighbors are related.
- **jpg:**
Good neighbors: png, svg, gif, image, which are image formats.
Poor neighbors: szczepanek, buteo, galleria, regnum, fiav, which are unrelated and likely noise.
- **the:**
Good neighbors: in, of, a, other high-frequency function words.
Poor neighbors: its, which, entire, part, second, which are not true function-word equivalents.
- **google:**
Good neighbors: yahoo, webpage, web, search, com, blogging, which are internet-related.
Poor neighbors: faq, editable, archived, redirection, which are webpage terms but not central to “Google.”, just webpage-related.

Comparison across embedding dimensions:

- **20 dimensions:** Representations are somehow too coarse. Many unrelated terms appear, like, `dog` → `ass`, `bikini`; `jpg` → `colspan`, `ffffff`. These terms are more unrelated to the central semantic, sometimes are noises.
- **50 dimensions:** A better representation. Most neighbors are semantically related, though occasional noise remains.
- **100 dimensions:** This dimension captures much better semantic detail. For example, `seattle` → `intelligencer` (local newspaper); `dog` → rare breeds like `komondor`, `borzoi`; `communist` → `marxist`, `communism`; `google` → `gmail`, `pagerank`, `maps`. Noise still exists for rare tokens (e.g., in `jpg`), but overall quality is higher. It seems that the neighbors are more centralized to the chosen word.

The results above shows **better** semantic representation for **larger** dimensions.

- (b) **Answer.** We show the trivial label of humans thoughts after the expression, following the top-10 retrieved words (with **red** ones are bad, **blue** ones are bad).

For dimension 10:

- king - man + woman: **Desired: queen**
anjou flavia pepin desiderius kampaku quarrelled magister pretender highness chatillon
- paris - france + uk: **Desired: london**
npr fest bbc cbc afi weekly itv syndicated diversions tulip
- hitler - germany + italy: **Desired: mussolini**
fateful untenable misfortunes catiline bibulus caligula bellicose cicero treachery antony
- child - goose + geese: **Desired: children**
cohabiting maternity circumcising intercourse cruelty closeted marriageable midwife underage spouse
- goes - eats + ate: **Desired: went**
go sitting punting cheered loudly cheering undressed got glad gotta
- car - road + air: **Desired: airplane (an aircraft)**
sentry airdrop countermeasures aircrew torpedoes armoring airborne **aircraft** detonated expendable

For dimension 50:

- king - man + woman: **Desired: queen**
matilda **queen** throne carloman son melisende disgrace sibylla daughter betrothed
- paris - france + uk: **Desired: london**
odeon magazine monthly cafe promo npr retrospective lulu listings woodstock

- hitler - germany + italy: **Desired: mussolini**
mussolini cesare petacci innitzer banality honoria accomplices benito conspirators aetius
- child - goose + geese: **Desired: children**
children midwife infants parents pregnant adults mothers fathering castrated adult
- goes - eats + ate: **Desired: went**
going got went go wakes gone wondering sat coming snuck
- car - road + air: **Desired: airplane (an aircraft)**
parachutes aircraft jet usaaf stovl carrier interceptor rocket warhawk airplane

For dimension 200:

- king - man + woman: **Desired: queen**
queen marries isabella sibylla consort anjou vasa heiress valois betrothed
- paris - france + uk: **Desired: london**
london leeds molview glasgow reissue cafe bluerhinos promo gig premiered
- hitler - germany + italy: **Desired: mussolini**
mussolini himmler rome loos speer petacci naples nazi cesare ribbentrop
- child - goose + geese: **Desired: children**
children infants parents mothers sexually young pups parenting infanticide midwife
- goes - eats + ate: **Desired: went**
went got going go coming gets gone walked came wakes
- car - road + air: **Desired: airplane (an aircraft)**
helicopter aircraft pressurized airframe psa interceptor usaf curtiss usmc parachutes

What works well vs. poorly

- *Works well at most cased of large dimensions (200d):*
gender/royalty (king-man+woman⇒queen), capital-country (paris-france+uk⇒london), leader-country pairs, pluralization (child-goose+geese⇒children), tense mapping (goes-eats+ate⇒went), and domain shifts to aviation for car-road+air.
- *Fails at lower dimensions:* lower dimensions fails to model complex concepts (it stands to reason that 10 dimensions cannot correctly capture 20 concepts linearly)

Analogy Example	Concept
king - man + woman → queen	Gender / Royal title (male ↔ female counterpart)
paris - france + uk → london	Capital-of relation (capital ↔ country)
hitler - germany + italy → mussolini	Leader-of relation (Nazi leader ↔ nation)
child - goose + geese → children	Pluralization (singular ↔ plural)
goes - eats + ate → went	Verb tense (present ↔ past)
car - road + air → airplane/aircraft	Transportation medium (land ↔ air)

Table 1: Underlying concepts.

- 10d cannot represent enough relations; nearest neighbors drift to topical or stylistic artifacts (e.g., npr, bbc for the paris analogy).
- Even at 50d, paris-france+uk veers into media/culture because paris and uk share heavy co-occurrence with outlets (npr/itv/magazine/promo) - Word2Vec captures semantics via co-occurrence. The resulting direction captures “Anglo media” rather than “capital-of(UK)”.
- For goes-eats+ate, high-frequency inflectional neighbors (e.g., going, got) can overweight the exact tense target at 50d.
- car-road+air lands on the broader aviation region; aircraft/helicopter are semantically valid but not the exact lexical target airplane. It shows that the variations will disperse the accuracy.

Why the arithmetic works

- **Linear representation hypothesis:** Many lexical relations correspond to approximately constant vector offsets (e.g., gender, tense, number). The vector king-man isolates a “royalty-minus-masculine” direction; adding woman transports that relation to the feminine neighborhood near queen.
There are interesting researches of *Linear representation hypothesis* in modren LLMs like Toy Models of Superposition. Other papers trying to understand such phenomenon, like On the Origins of Linear Represent
- **Corpus:** Wikipedia corpora bias vectors toward media, institutions, and high-frequency patterns, which may be the reason of the strong linkage of UK/Frence with co-ocured medias

Problem 9

Answer. (1) We cannot justified the Besag's approximation with Chain rule and naive backoff.

$$p(A, B, C, D) = p(A | B, C, D) p(B | C, D) p(C | D) p(D).$$

If we “back off” each conditional to only one neighbor along a path, we might approximate $p(A | B, C, D) \approx p(A | B)$, $p(B | C, D) \approx p(B | C)$, and keep $p(C | D)$ while leaving $p(D)$ as a marginal, which gives

$$p(A | B) p(B | C) p(C | D) p(D),$$

It is *not* Besag's $p(A | B) p(B | C) p(C | D) p(D | A)$: the last factor $p(D)$ is unconditional here.

(2) Fix: normalization We need to maintain the symmetric property as well as probability distribution. So we can add a normalizing constant ensuring the joint sums to 1, i.e

$$p(A, B, C, D) = \frac{1}{Z} p(A | B) p(B | C) p(C | D) p(D | A),$$

where $Z = \sum_{ABCD} p(A | B) p(B | C) p(C | D) p(D | A)$ Furthermore, considering the circular structure, each word may depends equally of its both sides. So we may model the probability as:

$$P(A, B, C, D) = \frac{1}{Z} p(A | B, D) p(B | A, C) p(C | B, D) p(D | C, A),$$

Problem 10

Proof. Using the conditional chain rule,

$$\begin{aligned} p(\neg\text{fortune}, \neg\text{race}, \neg\text{horse}, \neg\text{shoe} | \neg\text{nail}) &= p(\neg\text{fortune} | \neg\text{race}, \neg\text{horse}, \neg\text{shoe}, \neg\text{nail}) \\ &\quad \times p(\neg\text{race} | \neg\text{horse}, \neg\text{shoe}, \neg\text{nail}) \\ &\quad \times p(\neg\text{horse} | \neg\text{shoe}, \neg\text{nail}) \\ &\quad \times p(\neg\text{shoe} | \neg\text{nail}). \end{aligned}$$

From (a), $p(\neg\text{shoe} | \neg\text{nail}) = 1$. From (b) and Question 1(k) (if $p(W | Y) = 1$ then $p(W | Y, Z) = 1$), we have $p(\neg\text{horse} | \neg\text{shoe}, \neg\text{nail}) = 1$. From (c) and 1(k), $p(\neg\text{race} | \neg\text{horse}, \neg\text{shoe}, \neg\text{nail}) = 1$. From (d) and 1(k), $p(\neg\text{fortune} | \neg\text{race}, \neg\text{horse}, \neg\text{shoe}, \neg\text{nail}) = 1$. Therefore

$$p(\neg\text{fortune}, \neg\text{race}, \neg\text{horse}, \neg\text{shoe} | \neg\text{nail}) = 1.$$

Since the event $(\neg\text{fortune} \wedge \neg\text{race} \wedge \neg\text{horse} \wedge \neg\text{shoe}) \subseteq \neg\text{fortune}$ by Question 1(a) gives

$$1 = p(\neg\text{fortune}, \neg\text{race}, \neg\text{horse}, \neg\text{shoe} | \neg\text{nail}) \leq p(\neg\text{fortune} | \neg\text{nail}) \leq 1.$$

Hence $p(\neg\text{fortune} | \neg\text{nail}) = 1$

□